

Problem 6.11

Given $A(5) = \$7,500$ and $A(11) = \$9,000$. What is $A(0)$ assuming compound interest?

Under compound interest,

$$A(t) = A(0)(1 + i)^t,$$

where i is the effective annual interest rate.

From the given information,

$$7500 = A(0)(1 + i)^5,$$

$$9000 = A(0)(1 + i)^{11}.$$

Dividing the second equation by the first,

$$\frac{9000}{7500} = (1 + i)^6.$$

So,

$$1.2 = (1 + i)^6,$$

and therefore

$$1 + i = (1.2)^{1/6}.$$

Thus,

$$A(0) = \frac{7500}{(1 + i)^5} = \frac{7500}{(1.2)^{5/6}}.$$

Hence,

$$\boxed{A(0) = \frac{7500}{(1.2)^{5/6}} \approx 6442.8332}.$$